Meeting 4C — Runtime State Architecture

Part 3: Tabla, Tanpura, Metronome & LED State Machines

Purpose

This document defines the remaining high-level runtime state machines responsible for accompaniment engines and visual feedback.

These are **actual firmware states**. They represent modes that the ESP32 explicitly enters and exits during normal operation.

Runtime State Machine 5 — Tabla Engine

The tabla engine manages rhythm playback independently of the keyboard voice engine.

T1 — TABLA_IDLE

Purpose

Tabla engine inactive.

Entry

- System startup.
- Tabla stopped.

Responsibilities

- Consume no playback resources.
- Wait for user command.

Valid Transition

→ TABLA_LOADING

Exit

User starts tabla.

T2 — TABLA_LOADING

Purpose

Load selected taal into RAM.

Entry

Tabla started.

Responsibilities

- Locate selected taal.
- Load required samples.
- Initialize playback sequence.

Valid Transition

→ TABLA_READY

Exit

Resources loaded.

T3 — TABLA_READY

Purpose

Prepare rhythm playback.

Entry

Loading complete.

Responsibilities

- Initialize beat counter.
- Initialize tempo timer.
- Wait for Start.

Valid Transition

→ TABLA_RUNNING

Exit

Playback begins.

T4 — TABLA_RUNNING

Purpose

Continuously perform rhythmic accompaniment.

Entry

Playback started.

Responsibilities

- Advance beat position.
- Trigger bol samples.
- Maintain tempo accuracy.
- Loop through taal.

Valid Transition

→ TABLA_PAUSED

→ TABLA_STOPPING

Exit

Pause or Stop requested.

T5 — TABLA_PAUSED

Purpose

Suspend playback without resetting position.

Entry

Pause pressed.

Responsibilities

- Freeze beat counter.

- Preserve playback state.

Valid Transition

→ TABLA_RUNNING

→ TABLA_STOPPING

Exit

Resume or Stop.

T6 — TABLA_STOPPING

Purpose

Terminate accompaniment.

Entry

Stop requested.

Responsibilities

- Stop timer.
- Clear playback queue.
- Reset beat position.

Valid Transition

→ TABLA_IDLE

Exit

Engine reset.

Runtime State Machine 6 — Tanpura Engine

The tanpura engine continuously generates drone accompaniment.

TP1 — TANPURA_IDLE

Purpose

Drone inactive.

Entry

System startup or stop.

Responsibilities

Remain silent.

Valid Transition

→ TANPURA_LOADING

Exit

Start requested.

TP2 — TANPURA_LOADING

Purpose

Load tanpura sample set.

Entry

Tanpura enabled.

Responsibilities

- Load drone samples.
- Configure looping.
- Apply transpose.

Valid Transition

→ TANPURA_RUNNING

Exit

Resources prepared.

TP3 — TANPURA_RUNNING

Purpose

Generate continuous drone.

Entry

Loading completed.

Responsibilities

- Maintain seamless looping.
- Apply volume.
- Maintain pitch.

Valid Transition

→ TANPURA_STOPPING

Exit

Stop requested.

TP4 — TANPURA_STOPPING

Purpose

Terminate drone playback.

Entry

Stop requested.

Responsibilities

- Fade out if required.
- Stop looping.
- Release resources.

Valid Transition

→ TANPURA_IDLE

Exit

Playback ended.

Runtime State Machine 7 — Metronome

The metronome provides timing guidance independent of the tabla engine.

M1 — METRONOME_IDLE

Purpose

Metronome disabled.

Entry

Startup or stop.

Responsibilities

Remain inactive.

Valid Transition

→ METRONOME_RUNNING

Exit

User enables metronome.

M2 — METRONOME_RUNNING

Purpose

Generate timing clicks.

Entry

Enabled.

Responsibilities

- Maintain BPM.
- Trigger click sample.
- Count beats.
- Update beat indicator.

Valid Transition

→ METRONOME_PAUSED

→ METRONOME_STOPPING

Exit

Pause or Stop.

M3 — METRONOME_PAUSED

Purpose

Suspend timing.

Entry

Pause requested.

Responsibilities

Freeze beat counter.

Valid Transition

→ METRONOME_RUNNING

→ METRONOME_STOPPING

Exit

Resume or Stop.

M4 — METRONOME_STOPPING

Purpose

Terminate metronome.

Entry

Stop requested.

Responsibilities

- Reset beat counter.
- Stop timer.

Valid Transition

→ METRONOME_IDLE

Exit

Stopped.

Runtime State Machine 8 — LED Controller

The LED subsystem operates independently from the keyboard scanning logic.

L1 — LED_IDLE

Purpose

Normal idle illumination.

Entry

Startup.

Responsibilities

Maintain default LED state.

Valid Transition

→ LED_KEY_FEEDBACK

→ LED_PLAYBACK

→ LED_BOOT_TEST

Exit

New LED mode requested.

L2 — LED_BOOT_TEST

Purpose

Verify LED hardware.

Entry

During boot sequence.

Responsibilities

- Illuminate all LEDs.
- Verify LED driver.
- Clear LEDs.

Valid Transition

→ LED_IDLE

Exit

Test completed.

L3 — LED_KEY_FEEDBACK

Purpose

Display keys currently pressed.

Entry

Keyboard activity detected.

Responsibilities

- Turn corresponding LEDs ON.
- Turn LEDs OFF on release.

Valid Transition

→ LED_IDLE

Exit

No active notes.

L4 — LED_PLAYBACK

Purpose

Guide the musician during playback.

Entry

Playback begins.

Responsibilities

- Illuminate recorded notes.
- Follow playback timing.
- Extinguish notes after release.

Valid Transition

→ LED_IDLE

Exit

Playback completed.

L5 — LED_ERROR

Purpose

Indicate system fault.

Entry

Critical runtime error.

Responsibilities

- Blink warning pattern.
- Remain active until cleared.

Valid Transition

→ LED_IDLE

Exit

Error acknowledged.

Summary

This document introduces **19 additional firmware runtime states**.

Tabla

- TABLA_IDLE
- TABLA_LOADING
- TABLA_READY
- TABLA_RUNNING
- TABLA_PAUSED
- TABLA_STOPPING

Tanpura

- TANPURA_IDLE
- TANPURA_LOADING
- TANPURA_RUNNING
- TANPURA_STOPPING

Metronome

- METRONOME_IDLE
- METRONOME_RUNNING
- METRONOME_PAUSED
- METRONOME_STOPPING

LED Controller

- LED_IDLE
- LED_BOOT_TEST
- LED_KEY_FEEDBACK
- LED_PLAYBACK
- LED_ERROR

These state machines complete the major accompaniment and user-feedback subsystems. They remain independent of the synthesizer voice engine and communicate only through events generated by the central Runtime Manager.